

First Order Linear Differential Equations

Problem Set 3

PSet Tips

- Remember to explain the reasoning behind all of your answers.
- Keep your work organized and assign a page to each problem on Gradescope.
- You don't have to do this by yourself — office hours, the MQC, and your fellow students are great resources if you get stuck.

In problems 1 and 2, find the general solution to the differential equations.

1. $(x^2 + 1)\frac{dy}{dx} + 2xy - 3x = 0$

2. $xy' + y = 4\sqrt{x}; \quad x > 0.$

In problems 3–7, solve the initial value problems.

3. $\frac{dy}{dx} = -2xy + 2x e^{(-x^2)}; \quad y(0) = 2.$

4. $xy' = 2y + x^3 \sin(x); \quad y(1) = 0.$

5. $(x^2 + 1)y' + 5xy = x; \quad y(0) = 2.$

6. $xy' + 4y = \frac{e^{-x}}{x^3}; \quad y(1) = 0, x > 0.$

7. $(1+x)y' + y = \cos(x); \quad y(0) = 1, 1+x > 0.$